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MTGeophysics.jl: A software repository for magnetotelluric research and applications

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Software

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Summary

The magnetotelluric (MT) method uses natural electromagnetic field variations to image electrical resistivity tens to hundreds of kilometres into the Earth. Resistivity is diagnostic of the constituents that matter most for resources and hazards: saline fluids, partial melt, graphite, and metallic sulphides. MT therefore underpins exploration for critical minerals and geothermal energy, characterisation of aquifers, CO_2 storage sites and repository host rocks, and studies of volcanic and tectonic systems. That sensitivity, however, is ambiguous: a conductor at depth may be brine, melt, or ore. Conductance also trades off against depth, so many different subsurface configurations explain the same observations equally well. Standard practice nonetheless reports a single "best" model, with no measure of which imaged features the data require and which merely survive the modeller's choice of regularisation.

MTGeophysics.jl is a Julia package for MT modelling, inversion, and interactive visualization built around this problem. It provides the usual building blocks (1D and 2D forward solvers, ModEM 3D I/O (Kelbert et al., 2014), and interactive 3D viewers), but is organised around a single guiding idea: making stochastic inversion and uncertainty quantification practical for MT. The package is designed to move MT interpretation along the spectrum from one deterministic model, to an ensemble of plausible models, toward the full posterior distribution (Figure 1).

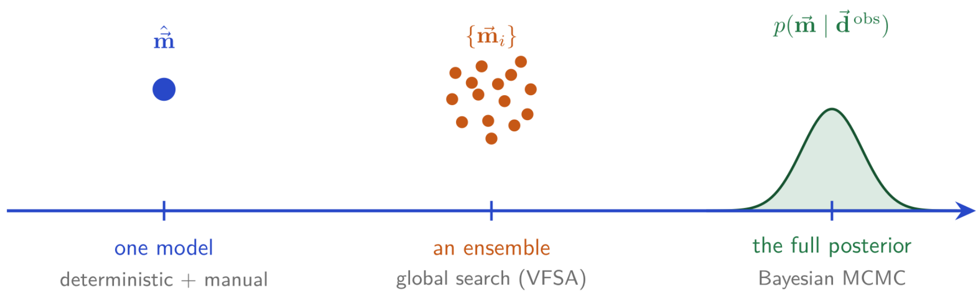

Figure 1: The vision behind MTGeophysics.jl. Deterministic inversion returns a single model $\hat{\mathbf{m}}$ chosen by a regularisation weight λ that the user picks; the regulariser expresses a *preference*, not information, so the "best" model shifts as λ changes. Greater rigour in uncertainty quantification (a VFSA ensemble $\{\mathbf{m}_i\}$, ultimately the full posterior $p(\mathbf{m} \mid \mathbf{d}^{\mathrm{obs}})$) costs more computation. The package is built to make the ensemble regime routine at regional scale and to open a practical path toward the full posterior.

Statement of need

Standard 3D MT inversion answers non-uniqueness by regularisation: it returns the single model minimising data misfit plus a smoothness penalty whose weight the user picks. The alternative is to sample many models consistent with the data. In 3D this has been considered impractical, because a regional mesh carries millions of free parameters and every candidate model requires a full electromagnetic forward solve.

MTGeophysics.jl closes this gap in practice. It provides a stochastic inversion workflow based on Very Fast Simulated Annealing (VFSA) (Sen & Stoffa, 2013) that produces an *ensemble* of plausible 3D models, from which ensemble statistics (mean, median, standard deviation) give a per-voxel picture of what the data do and do not constrain. The workflow reuses the community-standard ModEM forward solver (Egbert & Kelbert, 2012; Kelbert et al., 2014) by reading and writing its native file formats, so it slots into established MT practice without asking users to change their models, data, or solver.

The package targets MT researchers and students who want an open-source toolkit that builds on Julia's (Bezanson et al., 2017) strengths in numerical computing, composability, and interactive graphics. It is designed as a research repository and as a core component of a broader JuliaGeophysics ecosystem: forward solvers, data structures, and inversion routines are meant to be reused and recombined, so that new ideas can be prototyped without rebuilding core MT tooling. That composability also positions the package for the scientific-machine-learning direction set out below, and its plain-text formats and argument-driven example scripts keep every workflow reproducible from the command line and drivable by external tooling.

State of the field

Several open-source tools address parts of the MT workflow. ModEM (Egbert & Kelbert, 2012; Kelbert et al., 2014) is the community standard for 3D MT inversion but provides no built-in visualization or 1D/2D forward capabilities. MARE2DEM (Key, 2016) focuses on 2D and 3D marine electromagnetic modelling but is Fortran-based and limited to controlled-source methods. MTpy (Kirkby et al., 2019) provides comprehensive Python utilities for MT data handling and visualization but no forward solvers or stochastic inversion. pyGIMLi (Rücker et al., 2017) and SimPEG (Cockett et al., 2015) are general Python inversion frameworks with MT modules, but their MT functionality is embedded in much larger codebases. We are not aware of one that integrates stochastic inversion with ensemble uncertainty quantification in both 2D and 3D. MTGeophysics.jl is distinguished by the abstraction its components are organised around. A reduced model parameterisation makes ensemble inversion affordable at regional scale, and the package reuses the forward solver the MT community already trusts.

Software design

The central design choice in MTGeophysics.jl is reduced model parameterisation, the strategy that makes stochastic 3D inversion tractable. Stochastic inversion becomes intractable if every mesh cell is a free parameter: the search space is too large and each proposal requires an expensive forward solve. Rather than perturbing the full grid, the VFSA workflow perturbs a small set of radial-basis-function (RBF) control points and maps them back to the mesh with a compactly supported Gaussian RBF (truncated at 3σ). Each iteration reduces the padded modelling mesh to its core, samples M random control points, perturbs them by the VFSA rule, and interpolates back to the full mesh for the ModEM forward solve (Figure 2). Inverting for a handful of coefficients is what makes ensemble exploration feasible at regional scale.

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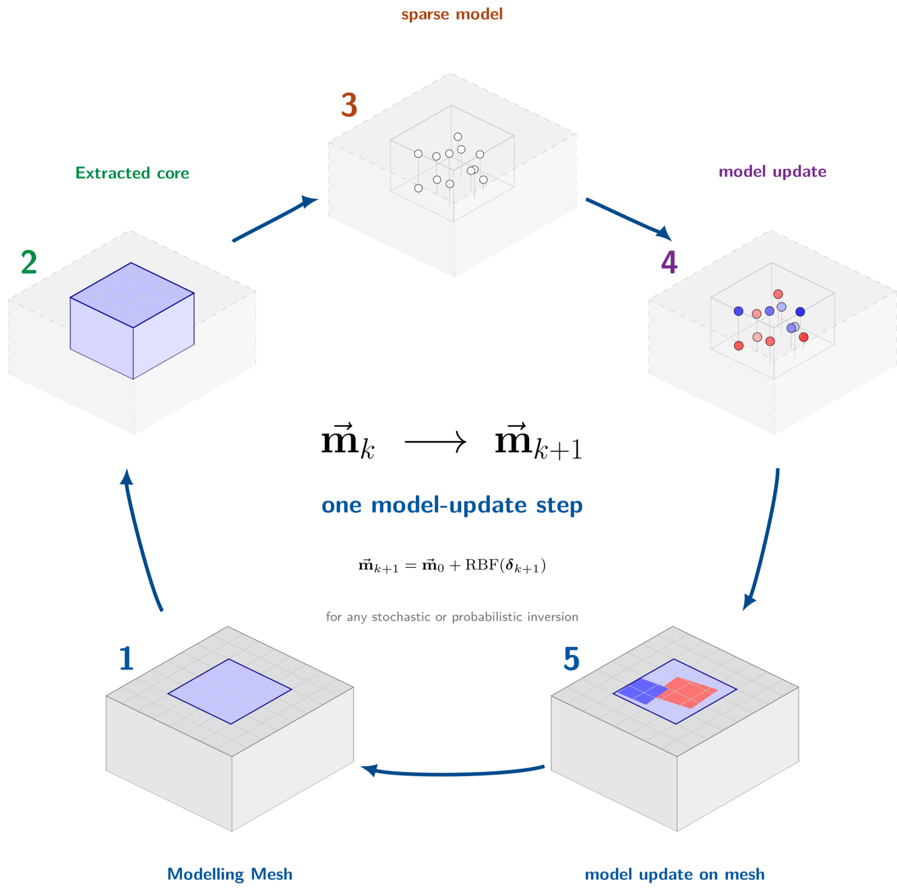

Figure 2: One VFSA iteration as a sparse-model update loop. The padded modelling mesh (1) is reduced to its core (2), sampled at M random control points, the *sparse model* (3), perturbed by the VFSA rule (4), then mapped back to the full padded mesh by compactly supported Gaussian-RBF interpolation (5) for the ModEM forward solve, advancing $\mathbf{m}_k \to \mathbf{m}_{k+1}$. This reduced parameterisation is the strategy that makes stochastic 3D inversion tractable.

The same principle extends beyond VFSA: searching a low-dimensional representation is a prerequisite for practical probabilistic inversion. Fully Bayesian approaches such as Markov chain Monte Carlo mirror the multi-chain structure already used here, but become affordable in 3D only under a similarly compact model. The roadmap follows the same logic: differentiable open-source forward solvers, neural surrogates for the expensive 3D solve, and implicit neural representations that recast the reduced parameterisation as a learned continuous field. That combination has already been demonstrated for 3D gravity inversion (Mishra, Laaksonen, et al., 2026), and Julia's scientific-machine-learning ecosystem makes MT the natural next target.

Around this core, the package is organised as a single Julia module with clearly separated layers: ModEM 3D data/model I/O; a 1D module with analytical and finite-difference solvers; a 2D module that assembles sparse finite-difference operators for the TE/TM Maxwell equations and solves them with LinearSolve.jl; and the VFSA inversion module. In 2D the package includes its own finite-difference forward engine; in 3D it wraps the external ModEM solver, running multiple independent Markov chains in parallel and computing ensemble statistics across them. The inversion driver reaches that solver only through plain-text files and a subprocess call, so the forward engine is swappable. Extending this interface to open-source 3D MT solvers is planned, which would remove the one component users must currently obtain separately. Interactive visualization is handled through GLMakie (GPU-accelerated 3D slice viewers with coordinate reprojection via Proj.jl and optional shapefile overlays), while CairoMakie produces publication-quality static plots. The visualization layer is conditionally loaded, so the core package runs without OpenGL dependencies. All file formats are plain text for reproducibility and version control. Detailed usage (the interactive slice viewers and the polygon-based model editor) is documented in the package documentation (Mishra, 2026).

Correctness is guarded by continuous integration on GitHub Actions. Every push and pull request runs the full test suite on Linux across two Julia versions: the minimum release declared in `Project.toml` (1.10) and a pinned current release, so both the supported floor and the latest toolchain are exercised. Because GLMakie cannot precompile without a display, the workflow provisions a virtual framebuffer (Xvfb) so the visualization layer is exercised headlessly in a clean environment. The suite combines smoke tests of the core 3D data structures and ModEM I/O with regression checks of the 1D and 2D forward solvers, including the COMMEMI benchmark generators, so forward-solver results are re-verified on every change. Companion workflows build and deploy the documentation and keep dependency bounds current.

Benchmarks and validation

The package is validated at two levels. The 2D forward solver and VFSA inversion reproduce the community-standard COMMEMI 2D benchmark suite (Zhdanov et al., 1997), generated natively by `helpers/benchmarks_2D.jl` and reproducible from the repository without external downloads. At regional scale, the 3D VFSA inversion is benchmarked against USArray MT data from the Cascadia subduction zone (Patro & Egbert, 2008), a dataset extensively studied with the deterministic ModEM NLCG inversion (Mishra, Kamm, et al., 2026). The VFSA ensemble mean recovers the major conductive structures found by deterministic inversion, while the ensemble standard deviation provides a per-voxel uncertainty estimate that no single deterministic inversion can offer; high-variance regions flag where the data poorly constrain the model. The benchmark is a substantial computation: each non-greedy chain performs roughly 12,000 ModEM forward solves at about 13.5 s each, close to two days per chain, with five chains run in parallel. The slice-by-slice comparison of deterministic model, ensemble mean, and ensemble standard deviation is reproduced in the package documentation (Mishra, 2026).

Research impact statement

MTGeophysics.jl supports ongoing magnetotelluric research at the Geological Survey of Finland (GTK) and opens uncertainty-aware MT interpretation to the wider community. Its components are reusable building blocks for the broader JuliaGeophysics ecosystem.

The package and the research built on it have been presented to the geophysics and scientific-computing communities by Pankaj K. Mishra and Cédric Patzer, including at the EGU General Assembly 2026, the 27th Electromagnetic Induction Workshop (EMIW 2026), and JuliaCon 2026. MTGeophysics.jl was also used to participate in the MT3DINV-4 workshop (Memorial University of Newfoundland, 2025), a community 3D MT inversion benchmarking exercise in which many groups inverted a common real field dataset acquired over the Raglan mining district in northern Quebec, Canada.

AI usage disclosure

GitHub Copilot was used as a coding assistant during development of this software and in drafting this paper. All AI-generated code and text were reviewed, tested, and verified by the author for correctness.

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